Let Q' be the quiver Q with orientation modified by the element $s_q \dots s_{n-1}s_n$. We let $k_{(i)}$ be the representation of Q' having dimension vector α_i. Then we define

$$V = F_n^- F_{n-1}^- \dots F_q^- k_{(i)}.$$

This is an indecomposable representation of Q and

$$d(V) = \alpha.$$

□

Example 6.8.5. Let us demonstrate by example how reflection functors work. Consider the quiver D_4 with the orientation of all arrows towards the node (which is labeled by 4). Start with the 1-dimensional representation V_{α_4} sitting at the fourth vertex. Apply to V_{α_4} the functor $F_3^- F_2^- F_1^-$. This yields

$$F_1^- F_2^- F_3^- V_{\alpha_4} = V_{\alpha_1+\alpha_2+\alpha_3+\alpha_4}.$$

Now applying F_4^-, we get

$$F_4^- F_1^- F_2^- F_3^- V_{\alpha_4} = V_{\alpha_1+\alpha_2+\alpha_3+2\alpha_4}.$$

Note that this is exactly the inclusion of three lines into the plane, which is the most complicated indecomposable representation of the D_4 quiver.

6.9. Problems

Problem 6.9.1. Let Q_n be the **cyclic quiver** of length n, i.e., n vertices connected by n oriented edges forming a cycle. Obviously, the classification of indecomposable representations of Q_1 is given by the Jordan normal form theorem. Obtain a similar classification of indecomposable representations of Q_2. In other words, classify pairs of linear operators $A : V \to W$ and $B : W \to V$ up to isomorphism. Namely:

(a) Consider the following pairs (for $n \geq 1$):

(1) $E_{n,\lambda}$: $V = W = \mathbb{C}^n$, A is the Jordan block of size n with eigenvalue λ, $B = 1$ ($\lambda \in \mathbb{C}$).

(2) $E_{n,\infty}$: is obtained from $E_{n,0}$ by exchanging V with W and A with B.